

ARC_FilterWheel.dll

Description



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Introduction to ARC_FilterWheel.dll

The **ARC_FilterWheel.dll** is provided by Acton Research Corporation (ARC) to allow for simplified low-level control of up to 16 Acton Research Corp. external ARC Filter controllers. This DLL does not work with other ARC products. To control other Acton Research Corp. devices you need a different set of DLLs. Feel free to contact Acton research for more information on these DLLs.

The DLL is broken into the following three function groups:

1. **Communications:** These functions are used for setting up communications with the ARC instrument(s) attached.
2. **Information:** These functions provided basic information about Open ARC instruments. This information includes such things as; model and serial number. (Note that older filter controllers do not store the serial number electronically).
3. **Filters:** These functions provide control of the external ARC Filter controllers.

Working with the DLL is a multi-step process:

1. First identify the number of external ARC Filter controllers attached by calling `ARC_Search_For_Filter`. Each instruments enumerates starting with zero up to the number of instruments found minus one. `ARC_get_Filter_preOpen_Model` will return the model string in a variant for the enumerated value passed to it.
2. `ARC_Open_Filter` is then called to open the enumeration. The function will return false (result = 0) if it failed to open the enumeration.
3. The user then calls the functions he needs to use to operate the instrument. Once opened, a filterwheel port can be left open until the software exits.
4. When the user is done with the instrument (usually program exit) the user calls `ARC_Close_Filter` to close the port.





Data Types

Most functions return a 16-bit unsigned integer result. True is defined as not equal to zero. The function failed if true is not returned.

- In VB this is an integer
- In VCC this is an integer
- In Delphi this is a WordBool

Parameters passed to and from a function are of the following types:

Integer values are 32-bit integers

- In VB this is a long
- In VCC this is a long
- In Delphi this is an integer

Floating-point values are doubles. This is true for most languages.

Strings are wrapped inside of the variant data type. This allows compatibility with most programming languages

- In VB this is a variant
- In VCC this is a variant
- In Delphi this is an OLEVariant (OLEVariant does not work the same as variant, you must include variants in the uses clause on Delphi 6 and higher)

Boolean values are 16-bit integers defined as true if not equal to zero.

- In VB this is an integer
- In VCC this is an integer
- In Delphi this is a wordbool



Brief Function Descriptions

Communications

Monochromators are opened in one of two ways, using ARC_Search_For_Filter / ARC_Open_Filter or to directly open the COM port with ARC_Open_Filter_Port.

ARC_Search_For_Filter , function used to search for attached monochromators. With the exception of ARC_Ver, must be the first function called (and only called once).

ARC_get_Num_Found_Filter_Ports , returns the value last returned by **ARC_Search_For_Filter** , can be called multiple times.

ARC_Open_Filter function to open a filter controller.

ARC_Open_Filter_Port , function to open a filter controller by addressing a specific COM port.

ARC_Close_Filter , function to close an open filter controller.

ARC_Valid_Filter_Enum , function returns if a filter controller enumeration is both valid and the instrument is open.

ARC_get_Filter_preOpen_Model , returns the model string of an instrument not yet opened. Note that ARC_Search_For_Filter needs to be called prior to this call.

Direct Communications

ARC_Ver , function provides the DLL version. It is the only function that can be called at any time.

Caution

ARC_Send_CMD_To_Filter , this **Direct Communications function** is an internal function that provides direct communications with the instrument. Usage of this function should be avoided.

Information

ARC_get_Filter_Model , returns the model string from the instrument.

ARC_get_Filter_Serial , returns the serial number from the instrument.





Filter

Model Types

- Model Type -1 ... Unknown, not supported
- Model Type 0 ... Unknown, not supported
- Model Type 1 ... GIMS (reserved for, none currently supported by dll)
- Model Type 2 ... SP-150 (superseded by SP-2-150 (SPv6x) in 2003)
- Model Type 3 ... SPx00i (generic gen2 board) (superseded by SPv6x in 2003)
- Model Type 4 ... SPx004i (generic SD-2) (superseded by SD-3 in 2003)
- Model Type 5 ... SP-300i (superseded by SP-2-300i (SPv6x) in 2003)
- Model Type 6 ... SP-500i (superseded by SP-2-500i (SPv6x) in 2003)
- Model Type 7 ... SP-750i (superseded by SP-2-750i (SPv6x) in 2003)
- Model Type 8 ... INS-300 (serial number is < 42) (superseded by SPv6x based INS-300 in 2003)
- Model Type 9 ... VM-502i (VM-502 with SD-2) (superseded by SD-3 based VM-502 in 2003)
- Model Type 10 ... VM-504i (VM-504 with SD-2) (superseded by SD-3 based VM-504 in 2003)
- Model Type 11 ... VM-503 (VM-503 with SD-2) (superseded by SD-3 based VM-503 in 2003)
- Model Type 12 ... VM-505 (VM-505 with SD-2) (superseded by SD-3 based VM-505 in 2003)
- Model Type 13 ... VM-506 (VM-506 with SD-2) (superseded by SD-3 based VM-506 in 2003)
- Model Type 14 ... VM-507 (VM-507 with SD-2) (superseded by SD-3 based VM-507 in 2003)
- Model Type 15 ... VM-510 (VM-510 with SD-2) (superseded by SD-3 based VM-510 in 2003)
- Model Type 16 ... VM-511 (VM-511 with SD-2) (superseded by SD-3 based VM-511 in 2003)
- Model Type 17 ... VM-512 (VM-512 with SD-2) (superseded by SD-3 based VM-512 in 2003)
- Model Type 18 ... VM-521 (VM-521 with SD-2) (superseded by SD-3 based VM-521 in 2003)
- Model Type 19 ... VM-523 (VM-523 with SD-2) (superseded by SD-3 based VM-523 in 2003)
- Model Type 20 ... SP-275 (superseded by SP-300i in 1996)
- Model Type 21 ... SP-500 (SP-500 with SD-748) (superseded by SP-500i in 1996)
- Model Type 22 ... SP-750 (SP-750 with SD-748) (superseded by SP-750i in 1998)





Model Type 23 ... VM-502 (VM-502 with SD-748) (superseded by VM-502i in 1999)

Model Type 24 ... VM-504 (VM-504 with SD-748) (superseded by VM-504i in 1999)

Model Type 25 ... SD-748 (superseded by SD-2 in 1999)

Model Type 26 ... INS-150

Model Type 27 ... SPv6x (All SpectraPro produced since 2003,

Includes: SP-2-150, SP-2-300i, SP-2-500i, SP-2-750i, SD-3, InSight,
MS-2-150i, MS-2-300i, INS-300 serial numbers > 42)

Model Type 28 ... NCL (superseded by SpectraHub in 2003)

Model Type 29 ... Filter Controller

Model Type 30 ... Reserved for Sample Chambers

Model Type 31 ... Reserved (custom, not supported by dll)

Model Type 32 ... Reserved (custom, not supported by dll)

Model Type 33 ... SpectraHub

Functions

ARC_get_Filter_Present , returns if the instrument has an integrated filter wheel

ARC_get_Filter_Position , returns the current filter position

ARC_set_Filter_Position , sets the current filter position

ARC_get_Filter_Min_Pos , returns the min. filter position

ARC_get_Filter_Max_Pos , returns the max. filter position

ARC_Filter_Home , homes the filter wheel

ARC_get_Filter_preOpen_Model_int32 , returns the model type of an instrument not yet opened (see Model Types below). Note that **ARC_Search_For_Inst** needs to be called prior to this call.

ARC_get_Filter_preOpen_Serial_int32 , returns the serial number (as an integer) of a filterwheel not yet open.





Detailed Function Descriptions

Communications

ARC_Search_For_Filter , function used to search for attached Filter Wheel. With the exception of ARC_Ver, must be the first function called.

Delphi : function ARC_Search_For_Filter(out Num_Found : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Search_For_Filter Lib "ARC_FilterWheel.dll" (Num_Found As Long) As Integer

Num_Found : The number of Acton Research Instruments found and enumerated. Enumeration list starts with zero and ends with Num_Found minus one.

Result : True, if any instruments where found.

ARC_get_Num_Found_Filter_Ports , returns the value last returned by **ARC_Search_For_Filter**. Can be called multiple times.

Delphi : function ARC_get_Num_Found_Filter_Ports : integer; stdcall;

VB : Private Declare Function ARC_get_Num_Found_Filter_Ports Lib "ARC_FilterWheel.dll" As Long

Result : Returns the value last returned by **ARC_Search_For_Filter**.

ARC_Open_Filter , function to open a Filter Wheel

Delphi : function ARC_Open_Filter(Enum_Num : integer; out Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Open_Filter Lib "ARC_FilterWheel.dll" (ByVal Enum_Num As Long, Filter_Enum As Long) As Integer

Enum_Num : Number in the enumeration list to open.

Filter_Enum : The handle by which to address the Filter Wheel in all future calls. Can be validated with ARC_Valid_Filter_Enum.

Result : True, if it successfully opened the device.





ARC_Open_Filter_Port , function to open a Filter Wheel on a specific COM Port (ie. COM1, COM2, ect...)

Delphi : function ARC_Open_Filter_Port(Com_Num : integer; out Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Open_Filter_Port Lib "ARC_FilterWheel.dll" (ByVal Com_Num As Long, Filter_Enum As Long) As Integer

Com_Num : Com Port (COM1,COM2,ect.) to open.

Filter_Enum : The handle by which to address the Filter Wheel in all future calls. Can be validated with ARC_Valid_Filter_Enum.

Result : True, if it successfully opened the device.

ARC_Close_Filter , function to close an open Filter Wheel

Delphi : function ARC_Close_Filter(Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Close_Filter Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Result : True, if the operations was carried out successfully.

ARC_Valid_Filter_Enum , function returns if a Filter Wheel enumeration is both valid and the instrument is open. All functions call this function to verify that the requested action is valid.

Delphi : function ARC_Valid_Filter_Enum(Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Valid_Filter_Enum Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Result : True if Filter_Enum is a valid Filter Wheel that is open.





ARC_get_Filter_preOpen_Model , returns the model string of an instrument not yet opened. Note that ARC_Search_For_Filter needs to be called prior to this call. In the case of multiple Filter Wheel/Spectrographs attached, it allows the user to sort, which instruments are to be opened before opening them.

Delphi : function ARC_get_Filter_preOpen_Model(Enum_Num : integer; out Model_Var : olevariant) : wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_preOpen_Model Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long, Model_Var As Variant) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Model_Var : The model string of the unopened Filter Wheel.

Result : True if the operation was successful

ARC_Send_CMD_To_Filter , this is an internal function that provides direct communications with the instrument. Usage of this function should be avoided.

Delphi : function ARC_Send_CMD_To_Filter(Filter_Enum : integer; SendCMD : olevariant; out RCV_Var : olevariant; ms_Timeout : integer) : wordbool; stdcall;

VB : Private Declare Function ARC_Send_CMD_To_Filter Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long, ByVal SendCMD As Variant, RCV_Var As Variant, ByVal ms_Timeout As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

SendCMD : The text command to be sent to the instrument. Note that line termination is handled by the DLL.

RCV_Var : The reply from the instrument to the given command.

Ms_Timeout : How long the DLL should wait in milliseconds for an acknowledgement before returning

Result : True if the instrument did not reject the command or timeout.





ARC_Ver , this function provides the DLL version. It is the only function that can be called at any time.

Delphi : function ARC_Ver(out Major : integer; out Minor : integer; out Build : integer) : wordbool; stdcall;

VB : Private Declare Function ARC_Ver Lib "ARC_FilterWheel.dll" (Majorval As Long, Minorval As Long, Buildval As Long) As Integer

Major : Version Major Number

Minor : Version Minor Number

Build : Version Build Number

Result : True if the operation was successful





Information

ARC_get_Filter_Model , returns the model string from the instrument.

- Delphi :** function ARC_get_Filter_Model(Filter_Enum : integer;out Model_Var :
olevariant); wordbool; stdcall;
- VB :** Private Declare Function ARC_get_Filter_Model Lib "ARC_FilterWheel.dll"
(ByVal Filter_Enum As Long, Model_str As Variant) As Integer
- Filter_Enum :** The handle defined in ARC_Open_Filter by which the Filter Wheel is
to be addressed.
- Model_Var :** Returns the model string of the instrument.
- Result :** True if the operation was successful

ARC_get_Filter_Serial , returns the serial number from the instrument. Note that older filter
controllers did not electronically store the serial number. In the
case of the older controllers Serial_Var will not contain any text.

- Delphi :** function ARC_get_Filter_Serial(Filter_Enum : integer;out Serial_Var :
olevariant); wordbool; stdcall;
- VB :** Private Declare Function ARC_get_Filter_Serial Lib "ARC_FilterWheel.dll"
(ByVal Filter_Enum As Long, Serial_str As Variant) As Integer
- Filter_Enum :** The handle defined in ARC_Open_Filter by which the Filter Wheel is
to be addressed.
- Serial_Var :** Returns the serial number of the instrument as a string.
- Result :** True if the operation was successful



Filter

ARC_get_Filter_Present , returns if the instrument has a filter wheel . Note if
ARC_Valid_Filter_Enum returns true, this function will also
return true.

Delphi : function ARC_get_Filter_Present(Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_Present Lib "ARC_FilterWheel.dll"
(ByVal Filter_Enum As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is
to be addressed.

Result : True if an integrated filter wheel is present on the Filter Wheel.

ARC_get_Filter_Position , returns the current filter position

Delphi : function ARC_get_Filter_Position(Filter_Enum : integer; out Position :
integer): wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_Position Lib "ARC_FilterWheel.dll"
(ByVal Filter_Enum As Long, Position As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is
to be addressed.

Position : Returns the current filter wheel position.

Result : True if the filter wheel is present and the operation was successful

ARC_set_Filter_Position , sets the current filter position

Delphi : function ARC_set_Filter_Position(Filter_Enum : integer; Position : integer):
wordbool; stdcall;

VB : Private Declare Function ARC_set_Filter_Position Lib "ARC_FilterWheel.dll"
(ByVal Filter_Enum As Long, ByVal Position As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is
to be addressed.

Position : Position the filter is to be set to.

Result : True if the filter wheel is present, position is not outside of
ARC_get_Filter_Min_Pos / ARC_get_Filter_Max_Pos and the operation was
successful



ARC_get_Filter_Min_Pos , returns the min. filter position

Delphi : Function ARC_get_Filter_Min_Pos(Filter_Enum : integer; out Position : integer): wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_Min_Pos Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long, Position As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Position : Returns the minimum position possible with the filter wheel.

Result : True if the filter wheel is present and the operation was successful

ARC_get_Filter_Max_Pos , returns the max. filter position

Delphi :function ARC_get_Filter_Max_Pos(Filter_Enum : integer; out Position : integer): wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_Max_Pos Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long, Position As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Position : Returns the maximum position possible with the filter wheel.

Result : True if the filter wheel is present and the operation was successful.

ARC_Filter_Home , homes the filter wheel.

Delphi : function ARC_Filter_Home(Filter_Enum : integer): wordbool; stdcall;

VB : Private Declare Function ARC_Filter_Home Lib "ARC_FilterWheel.dll" (ByVal Filter_Enum As Long) As Integer

Filter_Enum : The handle defined in ARC_Open_Filter by which the Filter Wheel is to be addressed.

Result : True if the filter wheel is present, the home operation succeeded, and the operation was successful.



ARC_get_Mono_preOpen_Model_int32, returns the model type of an instrument not yet opened. Note that ARC_Search_For_Inst needs to be called prior to this call.

Delphi : function ARC_get_Mono_preOpen_Model_Int32(Enum_Num : integer; out Model_Int32 : integer) : wordbool; stdcall;

VB : Private Declare Function ARC_get_Mono_preOpen_Model_int32(ByVal Enum_Num As Long , Model_Int32 As Long) As Integer

Enum_Num : The index as defined by ARC_Search_For_Inst for the instrument addressed.

Model_Int32 : The model type as a integer

Model Type -1 ... Unknown, not supported

Model Type 0 ... Unknown, not supported

Model Type 1 ... GIMS (reserved for, none currently supported by dll)

Model Type 2 ... SP-150 (superseded by SP-2-150 (SPv6x) in 2003)

Model Type 3 ... SPx00i (generic gen2 board) (superseded by SPv6x in 2003)

Model Type 4 ... SPx004i (generic SD-2) (superseded by SD-3 in 2003)

Model Type 5 ... SP-300i (superseded by SP-2-300i (SPv6x) in 2003)

Model Type 6 ... SP-500i (superseded by SP-2-500i (SPv6x) in 2003)

Model Type 7 ... SP-750i (superseded by SP-2-750i (SPv6x) in 2003)

Model Type 8 ... INS-300 (serial number is < 42) (superseded by SPv6x based INS-300 in 2003)

Model Type 9 ... VM-502i (VM-502 with SD-2) (superseded by SD-3 based VM-502 in 2003)

Model Type 10 ... VM-504i (VM-504 with SD-2) (superseded by SD-3 based VM-504 in 2003)

Model Type 11 ... VM-503 (VM-503 with SD-2) (superseded by SD-3 based VM-503 in 2003)

Model Type 12 ... VM-505 (VM-505 with SD-2) (superseded by SD-3 based VM-505 in 2003)

Model Type 13 ... VM-506 (VM-506 with SD-2) (superseded by SD-3 based VM-506 in 2003)

Model Type 14 ... VM-507 (VM-507 with SD-2) (superseded by SD-3 based VM-507 in 2003)

Model Type 15 ... VM-510 (VM-510 with SD-2) (superseded by SD-3 based VM-510 in 2003)

Model Type 16 ... VM-511 (VM-511 with SD-2) (superseded by SD-3 based VM-511 in 2003)

Model Type 17 ... VM-512 (VM-512 with SD-2) (superseded by SD-3 based VM-512 in 2003)





Model Type 18 ... VM-521 (VM-521 with SD-2) (superseded by SD-3 based VM-521 in 2003)

Model Type 19 ... VM-523 (VM-523 with SD-2) (superseded by SD-3 based VM-523 in 2003)

Model Type 20 ... SP-275 (superseded by SP-300i in 1996)

Model Type 21 ... SP-500 (SP-500 with SD-748) (superseded by SP-500i in 1996)

Model Type 22 ... SP-750 (SP-750 with SD-748) (superseded by SP-750i in 1998)

Model Type 23 ... VM-502 (VM-502 with SD-748) (superseded by VM-502i in 1999)

Model Type 24 ... VM-504 (VM-504 with SD-748) (superseded by VM-504i in 1999)

Model Type 25 ... SD-748 (superseded by SD-2 in 1999)

Model Type 26 ... INS-150

Model Type 27 ... SPv6x (All SpectraPro's produced since 2003,
Includes: SP-2-150, SP-2-300i, SP-2-500i, SP-2-750i, SD-3, InSight, MS-2-150i, MS-2-300i, INS-300 serial numbers > 42)

Model Type 28 ... NCL (superseded by SpectraHub in 2003)

Model Type 29 ... Filter Controller

Model Type 30 ... Reserved for Sample Chambers

Model Type 31 ... Reserved (custom, not supported by dll)

Model Type 32 ... Reserved (custom, not supported by dll)

Model Type 33 ... SpectraHub

Result : True if the operation was successful.

ARC_get_Filter_preOpen_Model_int32 , returns the model type of an instrument not yet opened. Note that ARC_Search_For_Inst needs to be called prior to this call.

Delphi : function ARC_get_Filter_preOpen_Model_int32 (Enum_Num : integer; out Model_Int32 : integer) : wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_preOpen_Model_int32 (ByVal Enum_Num As Long , Model_Int32 As Long) As Integer

Enum_Num : The index as defined by ARC_Search_For_Inst for the instrument addressed.

Model_Int32 : The model type as a integer

Model Type -1 ... Unknown, not supported

Model Type 0 ... Unknown, not supported





Model Type 1 ... GIMS (reserved for, none currently supported by dll)

Model Type 2 ... SP-150 (superseded by SP-2-150 (SPv6x) in 2003)

Model Type 3 ... SPx00i (generic gen2 board) (superseded by SPv6x in 2003)

Model Type 4 ... SPx004i (generic SD-2) (superseded by SD-3 in 2003)

Model Type 5 ... SP-300i (superseded by SP-2-300i (SPv6x) in 2003)

Model Type 6 ... SP-500i (superseded by SP-2-500i (SPv6x) in 2003)

Model Type 7 ... SP-750i (superseded by SP-2-750i (SPv6x) in 2003)

Model Type 8 ... INS-300 (serial number is < 42) (superseded by SPv6x based INS-300 in 2003)

Model Type 9 ... VM-502i (VM-502 with SD-2) (superseded by SD-3 based VM-502 in 2003)

Model Type 10 ... VM-504i (VM-504 with SD-2) (superseded by SD-3 based VM-504 in 2003)

Model Type 11 ... VM-503 (VM-503 with SD-2) (superseded by SD-3 based VM-503 in 2003)

Model Type 12 ... VM-505 (VM-505 with SD-2) (superseded by SD-3 based VM-505 in 2003)

Model Type 13 ... VM-506 (VM-506 with SD-2) (superseded by SD-3 based VM-506 in 2003)

Model Type 14 ... VM-507 (VM-507 with SD-2) (superseded by SD-3 based VM-507 in 2003)

Model Type 15 ... VM-510 (VM-510 with SD-2) (superseded by SD-3 based VM-510 in 2003)

Model Type 16 ... VM-511 (VM-511 with SD-2) (superseded by SD-3 based VM-511 in 2003)

Model Type 17 ... VM-512 (VM-512 with SD-2) (superseded by SD-3 based VM-512 in 2003)

Model Type 18 ... VM-521 (VM-521 with SD-2) (superseded by SD-3 based VM-521 in 2003)

Model Type 19 ... VM-523 (VM-523 with SD-2) (superseded by SD-3 based VM-523 in 2003)

Model Type 20 ... SP-275 (superseded by SP-300i in 1996)

Model Type 21 ... SP-500 (SP-500 with SD-748) (superseded by SP-500i in 1996)

Model Type 22 ... SP-750 (SP-750 with SD-748) (superseded by SP-750i in 1998)

Model Type 23 ... VM-502 (VM-502 with SD-748) (superseded by VM-502i in 1999)





Model Type 24 ... VM-504 (VM-504 with SD-748) (superseded by VM-504i in 1999)

Model Type 25 ... SD-748 (superseded by SD-2 in 1999)

Model Type 26 ... INS-150

Model Type 27 ... SPv6x (All SpectraPro's produced since 2003,
Includes: SP-2-150, SP-2-300i, SP-2-500i, SP-2-750i, SD-3,

InSight, MS-2-150i, MS-2-300i, INS-300 serial numbers > 42)

Model Type 28 ... NCL (superseded by SpectraHub in 2003)

Model Type 29 ... Filter Controller

Model Type 30 ... Reserved for Sample Chambers

Model Type 31 ... Reserved (custom, not supported by dll)

Model Type 32 ... Reserved (custom, not supported by dll)

Model Type 33 ... SpectraHub

Result : True if the operation was successful

ARC_get_Filter_preOpen_Serial_int32 , returns the serial number (as an integer) of a filterwheel not yet open.

Delphi : function ARC_get_Filter_preOpen_Serial_int32(Filter_Num : integer; out Serial_Int32 : integer) : wordbool; stdcall;

VB : Private Declare Function ARC_get_Filter_preOpen_Serial_int32 Lib "ARC_Instrument.dll" (ByVal Filter_Num As Long , Serial_Int32 As Long) As Integer

Filter_Num : The index as defined by ARC_Search_For_Inst for the instrument addressed.

Serial_Int32 : Returns the serial number as an integer.

Result : True if Filter_Num refers to a valid filterwheel.

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